08-05-2025 (M)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2025 –May 2025 Maximum Marks: 50 Examination: End-Semester Examination		Duration: 2 Hrs.
Programme code: 01 Programme: BTECH Computer	Class: LY	Semester: VIII (SVU 2020)
Name of the College: K. J. Somaiya School of Engineering	Name of the department: Computer Engineering	
Course Code: 116U01E864	Name of the Course: Natural Language Processing	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Identify and Justify the lexical relation between the highlighted words in sentences 1&2, 3&4, 5&6, 7&8 and 9&10. 1. He went to the **court** to fight the legal case. 2. The tennis match was played on the **court**. 3. She dropped a **bat** in the cave. 4. He swung the cricket **bat** with force. 5. The **bass** player led the band. 6. We saw a large **bass** fish in the lake. 7. The **soda** exploded when shaken. 8. He works for the **bottle** industry. 9. The **puppy** is very playful. 10. The **animal** was fed twice a day.	05
ii)	a) Tokenize the sentence: "The little boy quickly ran towards the big playground." b) Apply stemming to the following words: walking, studies, flying, beautifully, stronger (Use a common stemmer like Porter Stemmer.) c) Lemmatize the following words: playing, better, wolves, writing, drove d) Remove stop words from the sentence: "She is going to the market to buy some fresh apples	05
iii)	A news article and a poem are analyzed for TTR (Type Token ratio). Which one is expected to have a higher TTR and why?	05
Q.2	Attempt any **one**	10
i)	Define sentiment analysis and explain its significance in natural language processing. Discuss the main steps involved in performing sentiment analysis on text data.	
ii)	Write a note on Syntactic and Semantic Constraints on Coreference.	

Q.3	Attempt any **one**	10
i)	Explain with example Walker's Algorithm for word sense disambiguation.	
ii)	Explain with the help of example, step wise procedure for building Distributional Semantic Model (DSM).	
Q.4	Attempt the following	10

Consider the following PCFG fragment:

S	→	NP VP	1.0	NP	→	NP PP	0.4
VP	→	V NP	0.7	NP	→	*astronomers*	0.1
VP	→	VP PP	0.3	NP	→	*ears*	0.18
PP	→	P NP	1.0	NP	→	*saw*	0.04
P	→	*with*	1.0	NP	→	*stars*	0.18
V	→	*saw*	1.0	NP	→	*telescope*	0.1

Using the CKY algorithm, find the probability score for the most probable tree for sentence S = "**Astronomers saw stars with ears**" and draw all possible parse trees.

Q.5	Attempt the following	10

a) Write the algorithm for the minimum edit distance for spell correction using dynamic programming, which handles Insertion, Deletion, Substitutions, and Transposition.

b) Compute the minimum edit distance using dynamic programming between **'READING' & 'WRITING'**. (Assume both the insertion and deletion operations incur a cost of 1 each, while substitution incurs a cost of 2).